

LEAKLISTEN · REQUIREMENTS

LeakListen requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/03-requirements.md @ 20fe831		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3, status from LKL-CAL-001 for every requirement; R8 fixed (LKL-DDR-001 D6); R5 and R14 note the weekly time correction and the 24-byte summary
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

LeakListen requirements

These are first-pass requirements for one logger. Targets are proposals for review; they come from desk research, not yet from a partner utility, and must be revised after co-design (see the problem statement, LKL-PRB-001). Status is judged against the design in LKL-PRC-001 v0.3 by the calculations in LKL-CAL-001, which print every figure quoted here. The TRL 2 review items adopted for TRL 3 are recorded in LKL-DDR-001; no target is relaxed or redefined by them.

Table 1. Requirements and status at TRL 3 (LKL-CAL-001).

ID	Requirement	Target	Verification	Status at TRL 3
R1	Hear a leak on metallic mains	Flag a 5 L/min (1.3 US gal/min) leak at 3 bar within 100 m of the logger on cast or ductile iron DN100 to DN200	Attenuation and detection estimate (LKL-CAL-001 section G); later a partner site with a controlled leak	At risk: about 185 m on ductile iron at the central estimate, 93 to 370 m over plausible pipe losses
R2	Hear a leak on plastic mains	Same leak within 30 m on PVC or PE DN100 to DN200	As R1	Not met: about 4 m on PVC and 2 m on PE with a contact sensor on a valve
R3	Sensor bandwidth	5 Hz to 2 kHz, within ± 3 dB after correction	Resonance estimate (LKL-CAL-001 section F); later a shaker comparison	At risk: the magnet mount resonates at about 780 to 1,550 Hz, flat only to 420 to 840 Hz
R4	Sensor self-noise	Equivalent input noise 1 $\mu\text{g}/\text{VHz}$ or less from 100 Hz to 1 kHz	Noise calculation (LKL-CAL-001 section F)	At risk: 1.03 $\mu\text{g}/\text{VHz}$ worst case with the 53 g seismic mass
R5	Night listening	12 windows of 20 s between 02:00 and 04:00 local time, clock drift under 1 min per month	Clock drift estimate (LKL-CAL-001 section H)	Met on paper with a weekly network time correction (20 s worst); 87 s a month free running
R6	Battery life	5 years or more on one primary cell at the R5 schedule and one uplink a night	Power budget (LKL-CAL-001 section C)	Met on paper: 10.3 years or more on energy; taken as 10 years
R7	Radio link	90 % or more of nightly summaries delivered to a gateway within 1 km from a chamber with a cast-iron cover	Link budget (LKL-CAL-001 section D); later a field check	Not met: about 0.53 km under an iron cover at an assumed 20 dB cover loss (EU868 SF12)
R8	Data and privacy	Only band levels and spectra leave the logger; raw vibration samples are deleted on the device after processing. Fixed requirement (LKL-DDR-001 D6)	Design review	Met by design
R9	Attachment	Magnet holds 100 N or more on a steel or iron spindle cap, with no tools or pipe work	Hold estimate (LKL-CAL-001 section I); a pull test later	At risk: 149 N bare, 90 N on a 0.3 mm coating; not met for non-ferrous fittings
R10	Install from the surface	Placed and recovered in 10 min or less without entering the chamber, for chambers up to 1.5 m deep	Task estimate (LKL-CAL-001 section K); a walk-through with the partner utility	Not verifiable at TRL 3: 10 min estimate; cable reaches caps 2.09 m down
R11	Survive the chamber	IP68: 1 m submersion for 7 days; operate from -20 to +50 °C	Design review (LKL-CAL-001 section J)	Met by design; the logger floats on its lanyard when flooded
R12	Size and mass	Logger 70 mm diameter or less and 300 mm long or less; total mass 1.0 kg or less	Parametric model (LKL-CAL-001 section K)	Met on paper: 63 x 267 mm overall, 0.93 kg
R13	Leak flag quality			Not verifiable at TRL 3

ID	Requirement	Target	Verification	Status at TRL 3
		1 false alarm or fewer per 20 loggers per month after a 14-night baseline	Field data from a partner network	
R14	Open and interoperable	Standard LoRaWAN 1.0.x uplink (1.0.3 for the time request), documented 24-byte payload, works with any network server and TwinKit	Design review (LKL-CAL-001 sections A and B)	Met by design
R15	Cost	Parts cost \$130 or less per logger	Priced BOM (LKL-CAL-001 section M)	Met on paper: \$109.00

■ Requirements not met or at risk

Summary (LKL-CAL-001, Table 6): 2 not met, 4 at risk, 2 not verifiable at TRL 3, 4 met on paper and 3 met by design.

- **R2 not met.** Leak noise attenuates quickly on plastic pipes (0.42 dB/m at 100 Hz on PVC against 0.011 dB/m on ductile iron), and a contact sensor on a valve hears the reference leak only a few metres along PVC or PE. The hydrophone variant kept open in LKL-DDR-001 D3 is the likely answer; the scope for plastic networks is proposed in `docs/REVIEW.md`, awaiting Amish.
- **R7 not met.** At an assumed 20 dB cover loss the link closes to about 0.53 km, not 1 km. Options (proposed, awaiting Amish): a through-cover antenna, a composite cover, or a gateway within about 0.5 km of each district.
- **R1 at risk.** The central estimate of 185 m on iron clears 100 m, but pipe loss and leak source strength are both uncertain by large factors.
- **R3 at risk.** The magnet mount's resonance falls inside the band and varies by site.
- **R4 at risk.** 1.03 µg/VHz against 1 µg/VHz, on assumed ceramic and op-amp data.
- **R9 at risk** on coated ferrous caps, and not met on brass, bronze or plastic fittings, which need an adapter. A 42 mm magnet is proposed in `docs/REVIEW.md`.
- **R10 and R13 not verifiable at TRL 3.** They need a walk-through and field data with a partner utility.

■ Assumptions

- Network pressure at night 2 to 6 bar; leaks at lower pressure are quieter.
- Valve spindle caps and hydrant bodies are cast iron or steel on most networks the lab would target first.
- A LoRaWAN gateway (TwinKit, a partner's or a public network) is within about 1 km in an urban area. LKL-CAL-001 shows that under an iron cover it must be closer, about 0.5 km, unless the cover loss is lower than assumed.
- Transmit power follows the regional limit: +14 dBm in EU868, up to +20 dBm in US915.
- Primary cell: lithium thionyl chloride C size, 3.6 V, about 7.7 Ah nominal, derated to 70 % usable for cold and pulse loads.

■ Safety requirements

Safety: S1: no confined space entry is needed to place, service or recover a logger. S2: the cell is a primary lithium cell with a fuse or PTC and reverse protection; no charging circuit is fitted. S3: the logger carries a label with the owner, a contact and a lithium cell warning. S4: nothing touches drinking water, and nothing is fixed to the pipe other than the magnet.